Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CE723]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

ROCK MECHANICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	What episodes lead to start of ISRM? Identify and list out some specific areas of rock engineering applications.	**06**
	b.	Describe the distribution of rocks in Archean Groups. Explain the Cuddapah System of rocks.	**08**
2.	**a.**	Briefly explain Terzaghi's rock mass classification.	**07**
	b.	Based upon the six parameters briefly explain the RMR system.	**07**
3.	**a.**	Discuss the procedure for slake durability test with formula.	**09**
	b.	Explain the importance of Schmidt Hammer Test.	**05**
		OR	
4.	**a.**	Briefly explain the procedure to find the void ratio of rocks.	**07**
	b.	Explain the role of Sound Velocity in Rock Mechanics with a schematic diagram	**07**
5.	**a.**	Why is it necessary for Geotechnical engineer to understand stability of slopes in civil Engineering Constructions? Explain the classification of slope failure.	**07**
	b.	How do the plane failure occurs? Explain the favorable conditions of plane failure with a neat sketch	**07**
		OR	
6.	**a.**	Discuss the possible factors of slope failures. What are the objectives of slope stability analysis?	**06**
	b.	Explain toppling and buckling failure.	**08**
7.	**a.**	Briefly explain the foundation types on rocks.	**07**
	b.	Explain the IS guidelines for choosing the basis of design method for rock foundation.	**07**
		OR	
8.	**a.**	Briefly explain the estimation of bearing capacity of rocks	**08**
	b.	A dam is planned to be built near a town with Schists rock bed overlain by soft argillaceous conglomerate separated by a distinct fault. Its crushing strength was less than 4MPa. Conglomerate also did have veins of gypsum. As a Geotechnical engineer discuss on the reliability of data obtained and necessity for a field visit, in-situ tests and lab studies for further site investigation. Also give suggestion for construction worthiness of dam on the mineral composition found.	**06**